

Definite integrals – 2

The mean value theorems for the integral

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Theoretical minimum.

Theorem 1 (first mean value theorem). If f is continuous and g is integrable and does not change sign on $[a, b]$, then $\int_a^b fg = f(\xi) \int_a^b g$ for some $\xi \in (a, b)$. Without continuity of f one only gets a value between the infimum and the supremum of f .

Theorem 2 (second mean value theorem, Bonnet). If g is monotone and f is integrable on $[a, b]$, then there is $\xi \in [a, b]$ with $\int_a^b fg = g(a) \int_a^\xi f + g(b) \int_\xi^b f$. If moreover g is non-negative and non-increasing, then $\int_a^b fg = g(a) \int_a^\xi f$.

Fact 1 (what the second theorem is for). It converts a monotone factor into a boundary value, and so turns a pointwise bound on $\int f$ into a bound on $\int fg$: for monotone g one gets $|\int_a^b g(x) \sin(\lambda x) dx| \leq 4 \max |g|/\lambda$. This is the integral analogue of Abel's summation.

1. Prove both mean value theorems, show that in Theorem 1 the point ξ can always be taken in the open interval, and give an example of an integrable f for which no such ξ exists at all.
2. (RMO) Let $f: [a, b] \rightarrow \mathbb{R}$ be monotone and suppose that for all $x_1 < x_2$ there is $c \in (x_1, x_2)$ with $\int_{x_1}^{x_2} f = f(c)(x_2 - x_1)$.
 - a) Prove that f is continuous on (a, b) .
 - b) Does the conclusion survive if f is only assumed integrable?
3. (RMO) Let $f: [0, 1] \rightarrow \mathbb{R}$ be integrable with $0 < \int_0^1 f \leq 1$. Prove that there are $x_1 \neq x_2$ in $[0, 1]$ with $\int_{x_1}^{x_2} f = (x_1 - x_2)^{2002}$.

4. Prove the estimate of Fact 1 from the second mean value theorem, and deduce that $\int_0^1 g(x) \sin(\lambda x) dx \rightarrow 0$ for every monotone g , with an explicit rate. Show that without monotonicity no rate of this kind is possible.
5. (OMOUS 2022) Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous with $\int_0^1 f = \frac{\pi}{4}$. Prove that there exists $x_0 \in (0, 1)$ with $\frac{1}{1+x_0} < f(x_0) < \frac{1}{2x_0}$.